ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2025

Assignment 6 - Due date 02/27/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A06_Sp25.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: "ggplot2", "forecast", "tseries" and "sarima". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
#Load/install required package here

library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(tseries)
library(ggplot2)
library(Kendall)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr   1.1.4     v stringr 1.5.1
## v forcats 1.0.0     v tibble  3.2.1
## v purrr   1.0.2     v tidyr   1.3.1
## v readr   2.1.5

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
library(sarima)
```

```
## Loading required package: stats4
##
## Attaching package: 'sarima'
##
## The following object is masked from 'package:stats':
##
##     spectrum
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

This assignment has general questions about ARIMA Models.

Q1

Describe the important characteristics of the sample autocorrelation function (ACF) plot and the partial sample autocorrelation function (PACF) plot for the following models:

- AR(2)

  Answer: AR(2) is a second order autoregressive model. For AR models, the ACF will typically decay exponentially with time. The second order process indicates that the PACF has two significant lags. The order of AR models is dependent on the number of significant lags in the PACF plot.

- MA(1)

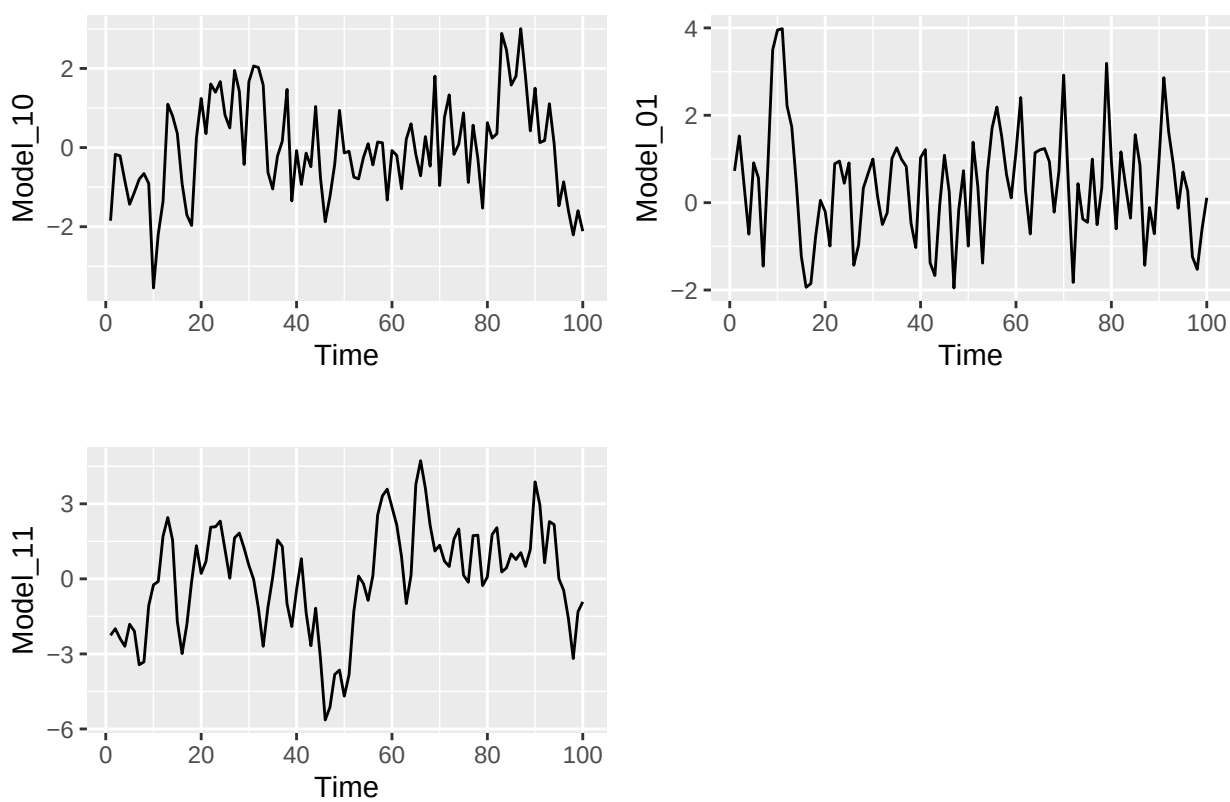
  Answer: MA(1) is a first order moving average model. For MA models, the ACF determines the order of the model. The first order process indicates that the ACF only has one significant lag. For MA models, the PACF will typically decay exponentially over time.

Q2

Recall that the non-seasonal ARIMA is described by three parameters ARIMA(p, d, q) where p is the order of the autoregressive component, d is the number of times the series need to be differenced to obtain stationarity and q is the order of the moving average component. If we don't need to difference the series, we don't need to specify the "I" part and we can use the short version, i.e., the ARMA(p, q).

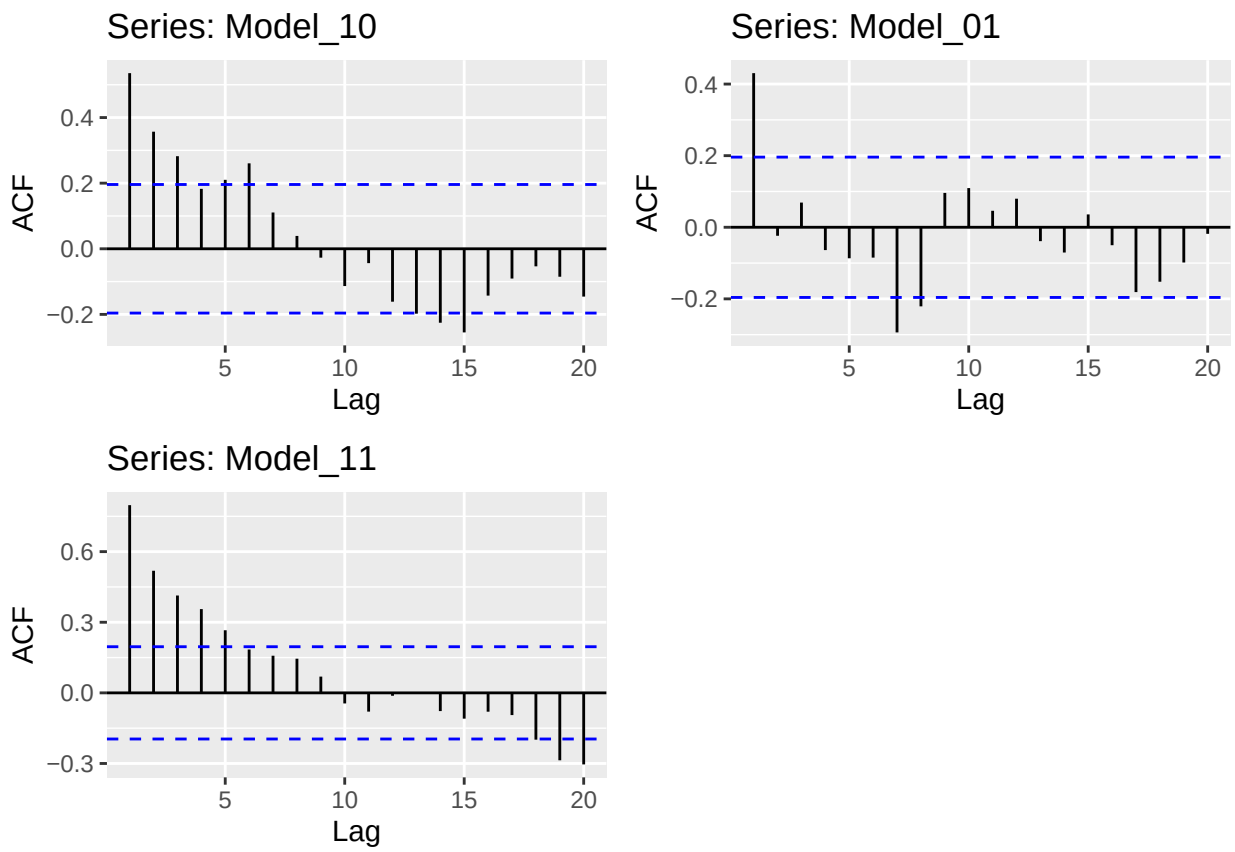
(a) Consider three models: ARMA(1,0), ARMA(0,1) and ARMA(1,1) with parameters $\phi = 0.6$ and $\theta = 0.9$. The ϕ refers to the AR coefficient and the θ refers to the MA coefficient. Use the `arima.sim()` function in R to generate $n = 100$ observations from each of these three models. Then, using `autoplot()` plot the generated series in three separate graphs.

```
Model_10 <- arima.sim(list(order = c(1,0,0), ar = 0.6), n = 100)


Model_01 <- arima.sim(list(order = c(0,0,1), ma = 0.9), n = 100)


Model_11 <- arima.sim(list(order = c(1,0,1), ar = 0.6, ma = 0.9), n = 100)


m10 <- autoplot(Model_10)
m01 <- autoplot(Model_01)
m11 <- autoplot(Model_11)

plot_grid(m10, m01, m11)
```



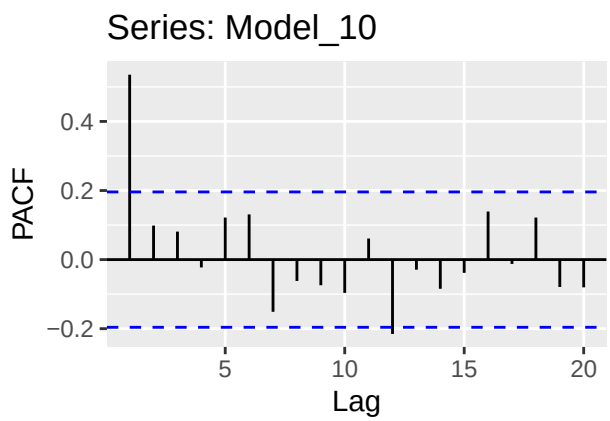
(b) Plot the sample ACF for each of these models in one window to facilitate comparison (Hint: use `cowplot::plot_grid()`).

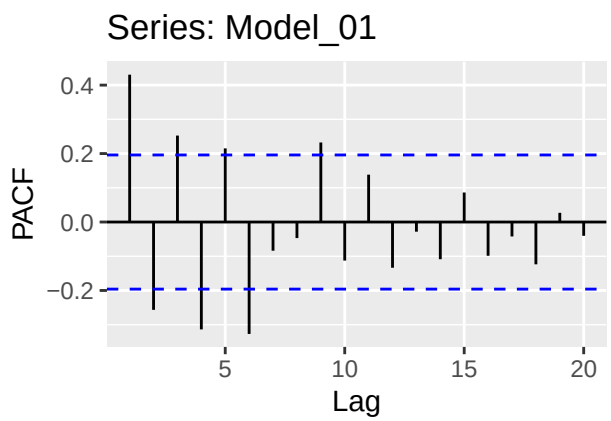
```
acf_m10<- autoplot(Acf(Model_10, plot = FALSE))
acf_m01 <- autoplot(Acf(Model_01, plot = FALSE))
acf_m11 <- autoplot(Acf(Model_11, plot = FALSE))

plot_grid(acf_m10, acf_m01, acf_m11)
```

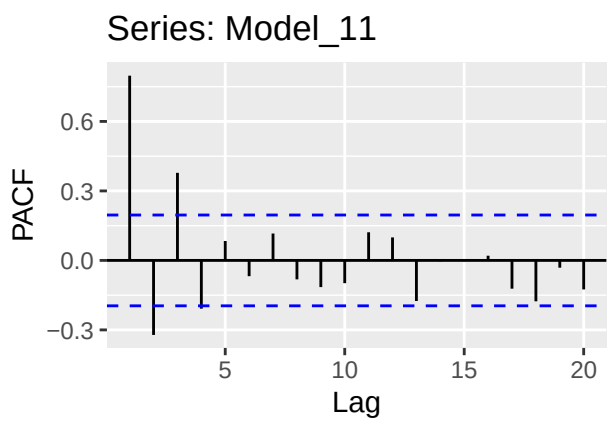


(c) Plot the sample PACF for each of these models in one window to facilitate comparison.

```
pacf_m10<- autoplot(Pacf(Model_10, plot = FALSE))
pacf_m01 <- autoplot(Pacf(Model_01, plot = FALSE))
pacf_m11 <- autoplot(Pacf(Model_11, plot = FALSE))

plot_grid(pacf_m10, pacf_m01, pacf_m11)
```







(d) Look at the ACFs and PACFs. Imagine you had these plots for a data set and you were asked to identify the model, i.e., is it AR, MA or ARMA and the order of each component. Would you be able identify them correctly? Explain your answer.

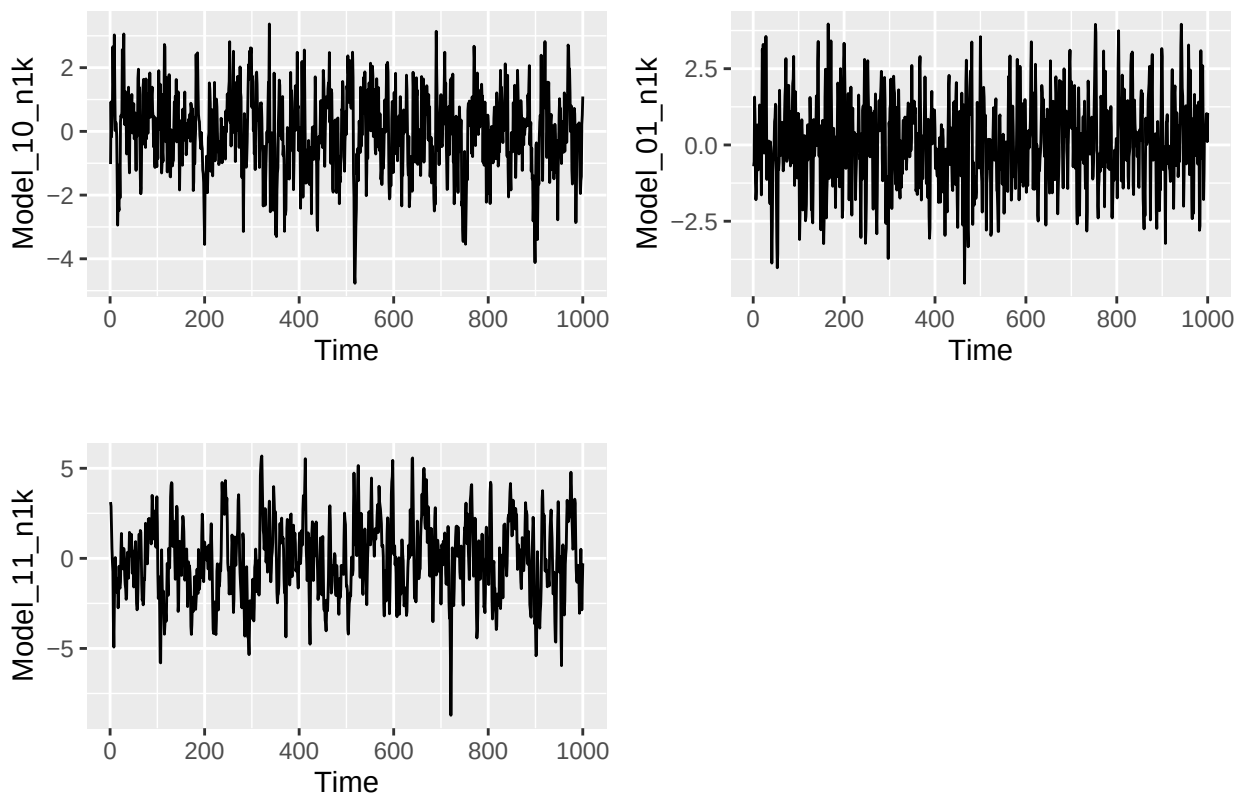
Answer: For the series Model_10 and Model_11, the exponential decay of the ACF indicates that the model has an AR component. The 1 significant lag in the PACF for Model_10 indicates that the series is a first order AR process. For Model_11, there are 2 significant lags in the PACF, however the PACF also appears to have an exponential decay pattern, indicating a potential MA component. For the series Model_01, the ACF cutting off and the exponential decay in the PACF indicates the series has an MA process. The 1 significant lag in the ACF of Model_01 indicates the series is a first order MA process. Truthfully, I may have been able to identify the Model_01 (MA(1)) and Model_10 (AR(1)) series correctly, however I most likely would have mis-identified Model_11 as ARMA(2,1).

(e) Compare the PACF values R computed with the values you provided for the lag 1 correlation coefficient, i.e., does $\phi = 0.6$ match what you see on PACF for ARMA(1,0), and ARMA(1,1)? Should they match?

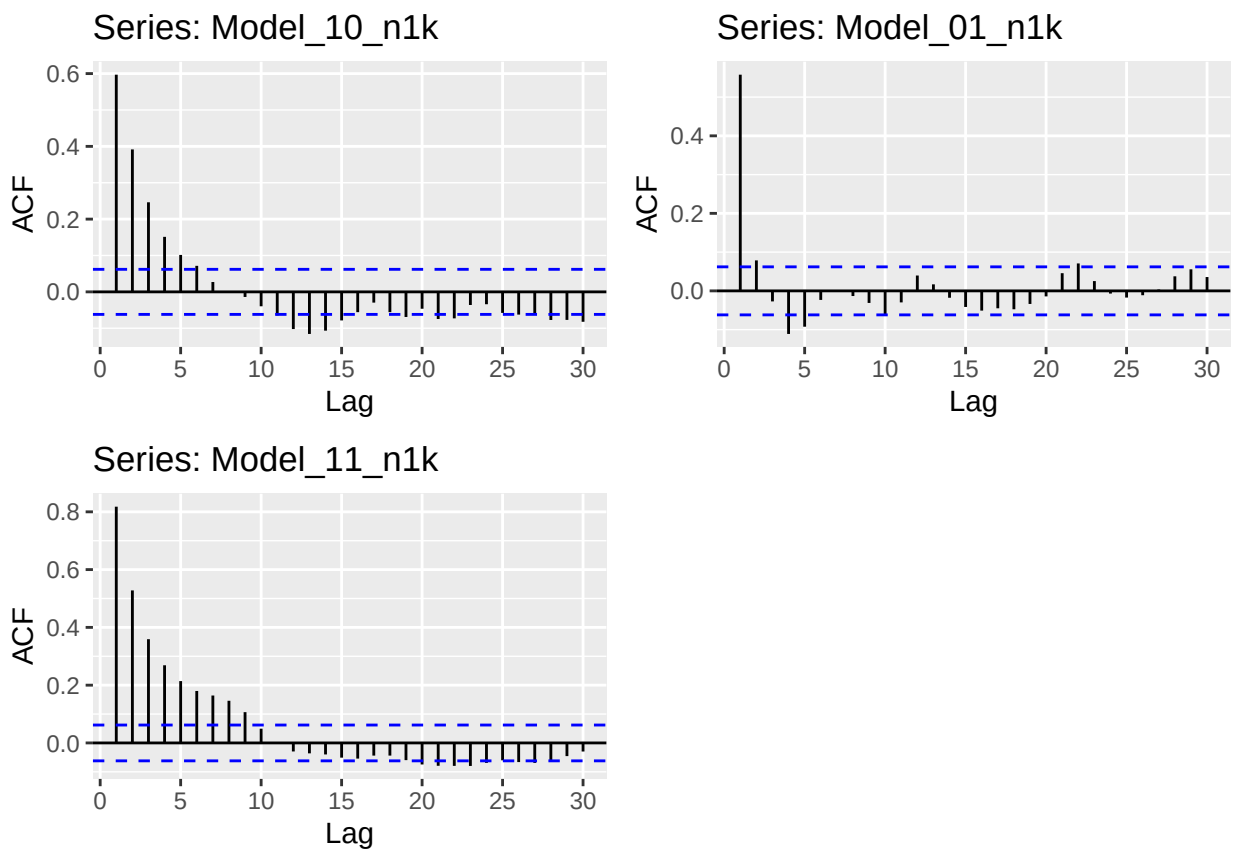
Answer: The PACF value for the lag 1 coefficient matched the value provided for the ARMA(1,0), but not the ARMA(1,1). This is because the MA component in the ARMA(1,1) most likely increased the correlation for the first lag in the PACF.

(f) Increase number of observations to $n = 1000$ and repeat parts (b)-(e).

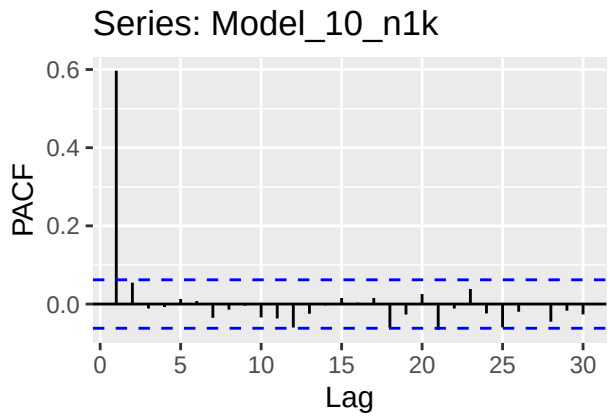
```r
Model_10_n1k <- arima.sim(list(order = c(1,0,0), ar = 0.6), n = 1000)


Model_01_n1k <- arima.sim(list(order = c(0,0,1), ma = 0.9), n = 1000)


Model_11_n1k <- arima.sim(list(order = c(1,0,1), ar = 0.6, ma = 0.9), n = 1000)

plot_grid(
  autoplot(Model_10_n1k),
  autoplot(Model_01_n1k),
  autoplot(Model_11_n1k))
```

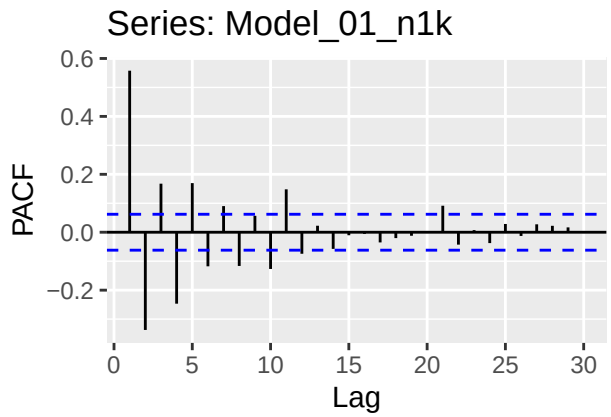


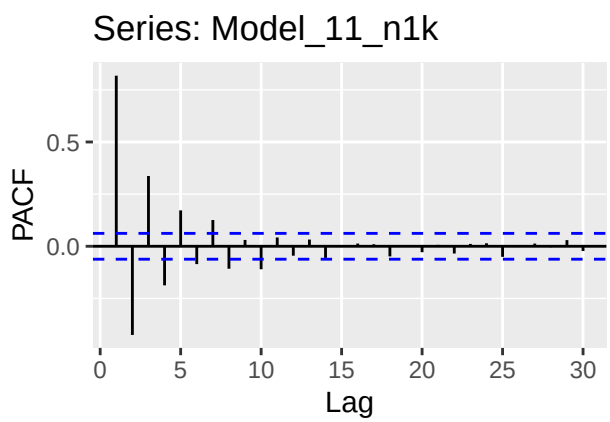

```r
#plotting the ACFs
plot_grid(
  autoplot(Acf(Model_10_n1k, plot = FALSE)),
  autoplot(Acf(Model_01_n1k, plot = FALSE)),
  autoplot(Acf(Model_11_n1k, plot = FALSE)))
```




```r
#plotting the PACFs
plot_grid(
  autoplot(Pacf(Model_10_n1k, plot = FALSE)),
  autoplot(Pacf(Model_01_n1k, plot = FALSE)),
  autoplot(Pacf(Model_11_n1k, plot = FALSE)))
```







Q3

Consider the ARIMA model $y_t = 0.7 * y_{t-1} - 0.25 * y_{t-12} + a_t - 0.1 * a_{t-1}$

(a) Identify the model using the notation ARIMA$(p, d, q)(P, D, Q)_s$, i.e., identify the integers p, d, q, P, D, Q, s (if possible) from the equation.

Answer 3a: ARIMA(1,0,1)(1,0,0)[12]

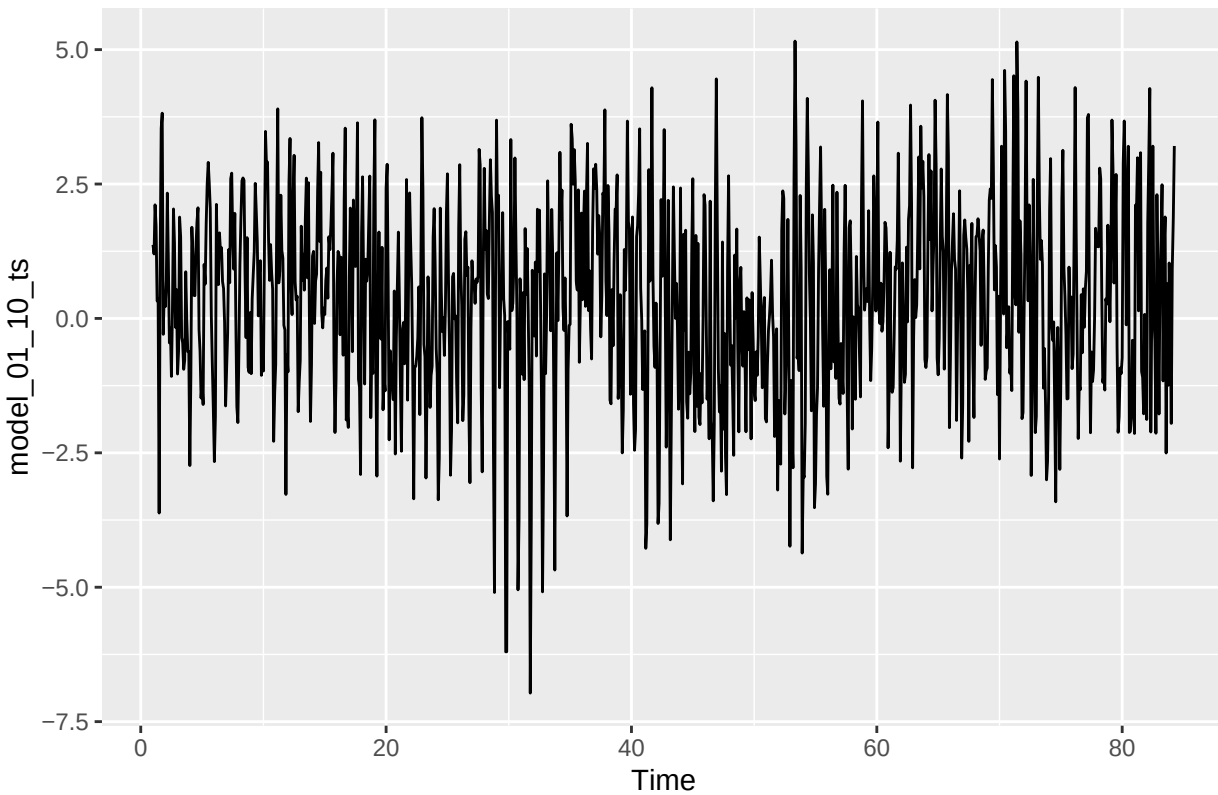
(b) Also from the equation what are the values of the parameters, i.e., model coefficients.

Answer 3b: The AR term coefficient value is 0.7, the MA term coefficient value is -0.1, and the SAR term coefficient value is -0.25.

Q4

Simulate a seasonal ARIMA$(0, 1) \times (1, 0)_{12}$ model with $\phi = 0.8$ and $\theta = 0.5$ using the `sim_sarima()` function from package `sarima`. The 12 after the bracket tells you that $s = 12$, i.e., the seasonal lag is 12, suggesting monthly data whose behavior is repeated every 12 months. You can generate as many observations as you like. Note the Integrated part was omitted. It means the series do not need differencing, therefore $d = D = 0$. Plot the generated series using `autoplot()`. Does it look seasonal?

```r
model_01_10 <- sim_sarima(
  model = list(ma = 0.5,
               sar = 0.8,
               nseasons = 12),n = 1000)

model_01_10_ts <- ts(model_01_10, frequency = 12)

autoplot(model_01_10_ts)
```

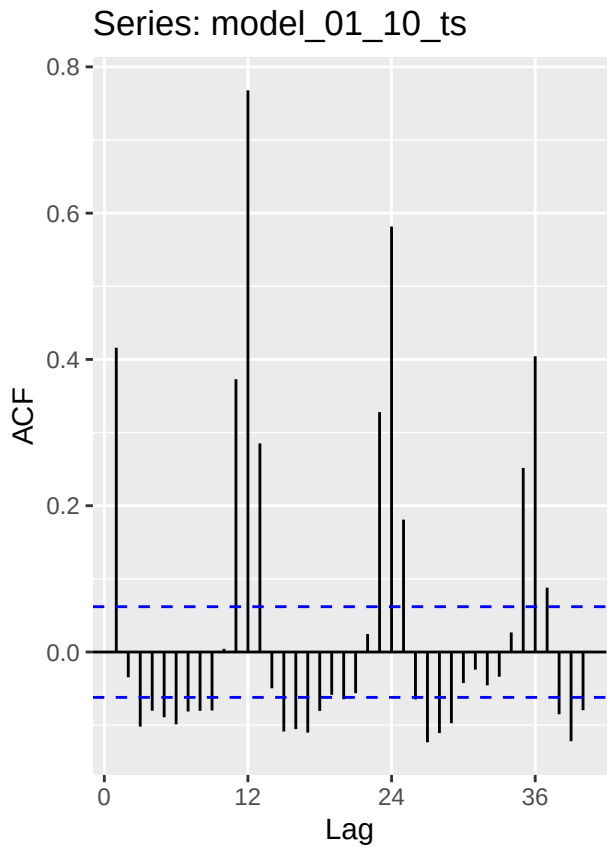


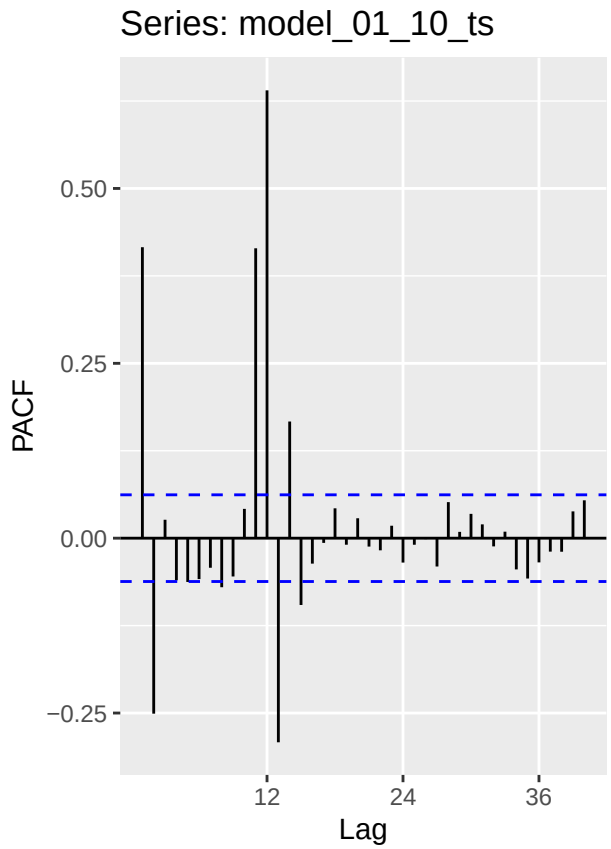
Answer: The plot does look seasonal. There is a general oscillating pattern in the plot that indicates the presence of seasonality.

Q5

Plot ACF and PACF of the simulated series in Q4. Comment if the plots are well representing the model you simulated, i.e., would you be able to identify the order of both non-seasonal and seasonal components from the plots? Explain.

```r
acf <- autoplot(Acf(model_01_10_ts, lag.max = 40, plot = FALSE))
pacf <- autoplot(Pacf(model_01_10_ts, lag.max = 40, plot = FALSE))

plot_grid(acf, pacf)
```





Answer: From the ACF and PACF plots of the simulated series, it would be possible to tell the order of the seasonal and non-seasonal components. Spikes in the ACF at 12, 24, and 36 as well as a single spike at lag 12 in the PACF indicate this series has an SAR term, or P = 1. The ACF cutting off instead of exponentially decaying indicates that the series has an MA term, and the single significant positive lag in the ACF indicates the order of the moving average component is first order, or q = 1.